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HATS: A Standard for the Hierarchical Adaptive Tiling Scheme in the Virtual Observatory

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Abstract

The increasing complexity and volume of astronomical datasets necessitate efficient spatial indexing and query strategies within the Virtual Observatory (VO). The Hierarchical Adaptive Tiling Scheme (HATS) is a framework designed to facilitate scalable queries, filtering operations, and efficient data retrieval across large astronomical surveys. Traditional spatial indexing methods often struggle with the massive scale of modern astronomical datasets, leading to inefficient query execution and storage overhead. HATS provides a flexible, hierarchical approach that balances computational efficiency and adaptability to non-uniform data distributions.

This document describes the structure, implementation, and best practices for integrating HATS within the VO ecosystem, ensuring interoperability and performance optimization for distributed astronomical datasets. We point out how HATS enhances existing indexing schemes, discuss its role in federated data access, and show how it enables powerful applications for large-scale survey science and efficient cross-matching of astronomical catalogs.

The reference implementation of HATS can be found at https://github.com/astronomy-commons/hats.

Status of this document

This is an IVOA Note expressing suggestions from and opinions of the authors. It is intended to share best practices, possible approaches, or other perspectives on interoperability with the Virtual Observatory. It should not be referenced or otherwise interpreted as a standard specification.

A list of current IVOA Recommendations and other technical documents can be found in the IVOA document repository[1].

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Conformance-related definitions

The words “MUST”, “SHALL”, “SHOULD”, “MAY”, “RECOMMENDED”, and “OPTIONAL” (in upper or lower case) used in this document are to be interpreted as described in IETF standard RFC2119 (Bradner, 1997).

The *Virtual Observatory (VO)* is a general term for a collection of federated resources that can be used to conduct astronomical research, education, and outreach. The International Virtual Observatory Alliance (IVOA) is a global collaboration of separately funded projects to develop standards and infrastructure that enable VO applications.

1 Introduction

The rapid expansion of astronomical data from large survey facilities like Vera C. Rubin Observatory, Euclid, and Roman Space Telescope necessitates innovative solutions for spatial indexing and efficient data retrieval. These surveys generate vast amounts of high-resolution imaging and time-domain data, requiring efficient methods for organizing, querying, and cross-

matching data across multiple archives. The challenges of storing and operating on imaging, spectral, and time-series data from multiple surveys are increasingly recognized (Nádvorník et al., 2021). Traditional approaches to spatial indexing, such as hierarchical pixelization (e.g., HEALPix Górski et al. (2005)) or static tiling schemes, often exhibit inefficiencies when handling uneven sky densities.

The Hierarchical Adaptive Tiling Scheme (HATS) is a novel approach designed to optimize spatial data partitioning while maintaining flexibility in accommodating varying data densities. Unlike fixed spatial partitioning methods, HATS adaptively adjusts tile sizes based on local data characteristics, ensuring an optimal balance between resolution, query efficiency, and storage management. By leveraging a hierarchical structure, HATS enables efficient spatial queries, allowing users to quickly focus on regions with fine-grained detail when needed, while maintaining scalability across large areas.

This document aims to define best practices for implementing and utilizing this tiling scheme within the Virtual Observatory framework. This document outlines the principles behind HATS, describes its data model, and provides recommendations. Additionally, we discuss how HATS can facilitate efficient cross-matching of astronomical catalogs, accelerate large-scale spatial queries, and enhance interoperability between diverse astronomical datasets.

2 Motivation and Goals

The primary motivation behind HATS is to address the following challenges in astronomical data management:

- **Scalability:** Modern astronomical surveys generate petabytes of spatially distributed data, requiring an indexing scheme that scales efficiently with dataset size. HATS provides such an indexing scheme, and creates files whose properties and size are well-suited to parallel operations.
- **Adaptive Spatial Resolution:** Fixed grid-based partitioning often leads to inefficient storage and query execution, particularly in non-uniformly distributed datasets. HATS adaptively adjusts tile sizes to accommodate varying data densities.
- **Efficient Query Execution:** Spatial queries such as nearest-neighbor searches and cross-matching must be executed efficiently across distributed data repositories. HATS enables rapid indexing and retrieval of relevant data subsets, and provides an balanced partitioning of data.
- **Interoperability:** Astronomical data is collected from diverse instruments and observatories, often using different spatial reference frames.

```
catalog/
|-- [REQUIRED] hats.properties
|-- [RECOMMENDED] partition_info.csv
|-- [OPTIONAL] skymap.fits
+-- dataset/
    |-- Norder=2/
    |-- Norder=3/
    |-- Norder=4/
    |-- Norder=5/
    |-- Norder=6/
    +-- Norder=. . ./
```

Figure 1: Example catalog directory contents

HATS provides a standardized framework for integrating and harmonizing spatial data across multiple repositories.

3 HATS Design and Implementation

3.1 HATS Catalog Directory Structure

The HATS framework relies on spatially sharding catalogs into Parquet files of approximately the same size. Here, we discuss how this is achieved and additional concepts that make it easier to use this main idea for astronomical research.

The central unit of data storage is the HATS `catalog`. It stores the data along with the associated metadata needed to access it. The `catalog` organization structure is shown in Figure 1.

The astronomy data is stored in the directory `dataset`, and further, within the subdirectories that specify the order at which particular part of the dataset is stored. We will discuss the partitioning and data storage in Sections 3.1.1, 3.1.2 and 3.1.3. The other files visible above are various metadata and auxiliary files that are here to enable better and easier handling of the data and we will describe them in Section 3.4.

3.1.1 Hierarchical Directory Structure

Focusing now on the dataset's contents, HATS employs a multi-level hierarchy based on HEALPix tiling. Each level corresponds to a single HEALPix order and represents a progressively finer spatial resolution.

All tiles of the same HEALPix order are contained within the same prefix `Norder=k` directory. To avoid directories becoming too large for some file systems, the tiles are then grouped by a `Dir` subdirectory prefix, where the value of the `Dir` key is the result of integer division by 10,000 of the pixel number.

```
dataset/
|-- . . .
|-- Norder=6/
|   |-- Dir=0/
|   |   |-- Npix=0.parquet
|   |   |-- . . .
|   |   +-- Npix=9999.parquet
|   +-- Dir=10000/
|       +-- Npix=10000.parquet
|           +-- . . .
|-- Norder=7/
+-- . . .
```

Figure 2: Example catalog dataset directory contents

```
dataset/
|-- . . .
|-- Norder=6/
|   |-- Dir=0/
|   |   |-- Npix=0/
|   |   |   |-- part0.parquet
|   |   |   +-- . . .
|   |   +-- Npix=9999/
|   |       |-- part0.parquet
|   |       +-- . . .
|   +-- Dir=10000/
|       +-- Npix=10000/
|           |-- part0.parquet
|           +-- . . .
|-- Norder=7/
+-- . . .
```

Figure 3: Example catalog dataset directory contents with leaf directories

We see the directory structure in Figure 2, showing a dataset with leaf Parquet files at several HEALPix orders.

The data is stored in the Parquet files (discussed in Section 3.1.3), with one or multiple files being possible in the final directory, i.e., in the ultimate data leaf.

If there are multiple files representing a single data partition, they should be read together, i.e., we consider them to be one single data unit. In this way, small updates can be added to already existing `catalogs` with simple, correctly placed, additions of files in existing folders.

With such functionality, it is possible to continuously update catalogs with small updates, e.g., nightly additions, without the need to rebuild the whole catalog. The price is the imbalance between the sizes of the data leaves, and the dataset needs to be rebuilt once this imbalance becomes too large from the standpoint of the catalog provider.

Such a directory structure would appear as shown in Figure 3.

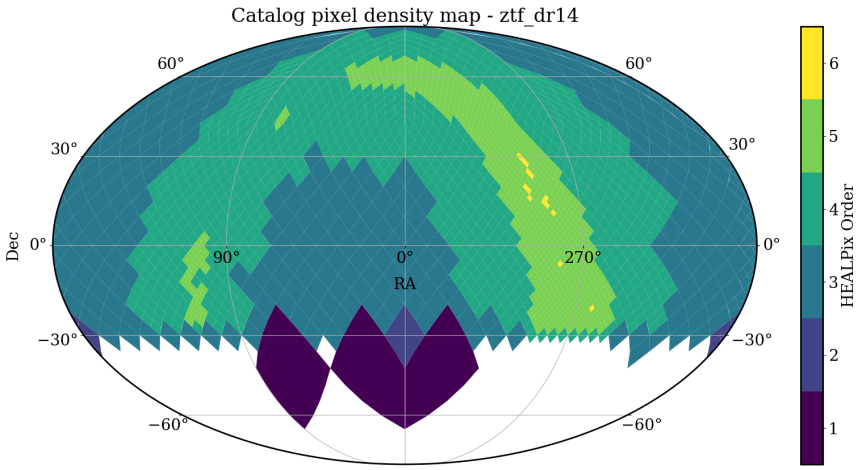


Figure 4: Adaptive tiling for the object catalog of Zwicky Transient Facility, Data Release 14 (Bellm et al., 2019; Masci et al., 2019) To achieve a similar number of rows across the entire sky, we partition the galactic region more finely. This results in tiles of higher HEALPix order in the galactic plane, particularly near the galactic bulge.

3.1.2 Adaptive Tiling Algorithm

Unlike static partitioning schemes, HATS adaptively subdivides spatial regions based on data angular density. In areas with sparse data, larger HEALPix tiles minimize storage overhead, whereas high-density areas are subdivided into smaller tiles to improve query efficiency.

The data is stored at a given level until the size of the data within the tile crosses a predetermined threshold. This threshold can be, most commonly, the number of rows or the size of the data on disk. As we add the data, at the point that the threshold is reached, the data gets split into four higher-order HEALPix tiles using the spatial information contained in the data. This process continues until all of the data is stored at the appropriate level and no data leaf has more data than the predetermined threshold. We show an example of the result of this procedure in Figure 4.

3.1.3 Structure of Data Files

The astronomical data is stored in Parquet format[2]. Parquet is a binary, columnar storage file format optimized for efficient data compression and retrieval. It is ideal for storing large amounts of astronomical tabular data because it enables fast access to specific columns without reading the entire dataset, significantly reducing I/O and improving performance. Due to its widespread integration with popular processing frameworks such as Dask,

[2] https://parquet.apache.org

```
hats_col_healpix       = healpix29
hats_col_healpix_order = 29
```

Figure 5: Example properties for the HEALPix spatial index column

Apache Spark, Hadoop, and cloud-native query engines, Parquet has become the de facto standard for analytical workloads in big data ecosystems. It is also widely adopted in cloud environments, such as Amazon S3 or Google Cloud Storage, and has libraries implemented for various programming languages, making it easy to work with Parquet files in different processing environments.

Additional optional performance considerations for Parquet files is discussed in Section 4.1.

3.1.4 Spatial Index

The HATS format RECOMMENDS that the first column of the dataset is a spatial index column. This column stores crucial spatial information about the position in the sky for each row, and it is not guaranteed to be unique. The existence of this value speeds up downstream spatial calculations, and is beneficial for spatially-intensive applications.

This MUST be is calculated from the row's right ascension (from the `hats_col_ra` column value) and declination (from the `hats_col_dec` column value). If two objects occur at the same location (or the data is individual observations of the same sky object), then multiple rows may have the same value for the column.

This document RECOMMENDS 29 as the HEALPix order used in the calculation, as it provides an incredibly granular index (HEALPix order 29 correspond to the mean separation between HEALPix pixels of 0.000393 arcsec), and is the largest such that will fit into a signed 64-bit integer. The name of the column and order at which it is calculated are specified in the `hats.properties` file (detailed in Section **??**), as seen in Figure 5.

3.2 Supplemental Tables

To improve scalability and efficient query optimization, HATS readers can benefit from alternative arrangements of the data. We refer to these as supplemental tables or supplemental catalogs, and these offer a limited or re-projected view of the primary dataset.

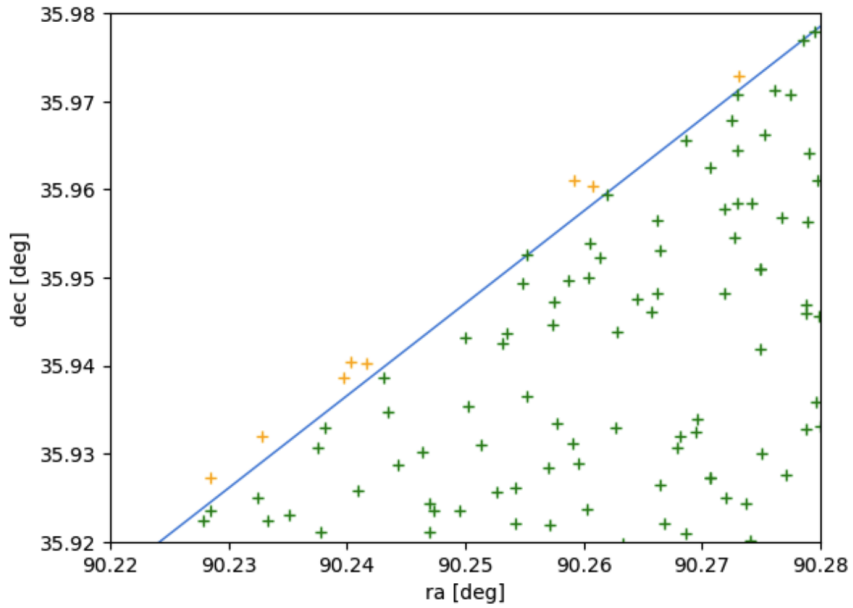


Figure 6: Example of margin contents. Green points are present in the primary catalog partition, the blue line is the HEALPix pixel boundary, and yellow points are points in the neighboring tile that are within 10 arcseconds of the boundary.

3.2.1 Margin Cache

An example of a supplementary table is the Margin Cache. One of the primary motivations for HATS is the rapid cross-matching of different surveys or different data sources within the same survey. The power of cross-matching in HATS is that each computation unit gets small spatially-connected parts from each dataset and works on them one part at a time, with the amount of data being roughly equal in each part, for best efficiency.

However, this introduces a limitation: at the boundaries of a divided section, some data points that should be cross-matched will be missed because they are present just across the border in neighboring partitions instead. This has a number of causes (observational error, non-point source positions, proper motion, etc), but it should be handled by applications to ensure the appropriate counterpart is found during cross-matching.

One option would be to load all neighboring partitions when performing a cross-match. However, this would require loading much unnecessary data, as only a small sliver around the edge of a partition is a candidate for counterparts in neighboring partitions. Figure 6 shows some example data points that are near to a HEALPix tile boundary, and should be included in a cross-match calculation.

We address this through the creation of a margin cache. This is an additional catalog that contains points in a limited angular threshold around the primary catalog partition. The margin catalog MUST contain all points within the indicated angular threshold from the tile boundary, but MAY contain additional points, either for simplicity of spatial calculations, or to provide additional points of interest.

The margin data can be loaded alongside the primary catalog partition data during a cross-match operation to account for any positional issues that may arise around the boundaries of the partition's HEALPix pixel. The margin catalog can be overloaded to also provide a cache of related objects that are not within the primary data partition for other scientific use cases.

3.2.2 Index Table

HATS catalogs are partitioned spatially, on right ascension and declination. This makes finding objects in a particular area of the sky very straightforward. However, one may occasionally only know the object by the survey-assigned identifier, and to find the row, would have to perform a full scan of all partitions.

HATS supports creating additional secondary index tables. To create index tables, we perform a full scan over all of the data, sort by the desired column, and write out a simple mapping from the unique identifier to where it can be found in the sky. Sorting over a large dataset can be expensive, and it's preferable to perform this operation once and store the results.

This mirrors the relational database notion of a lookup table, where you can lookup the spatial parameters of a non-spatial property. The user can query the index catalog for the HEALPix tile where the value can be found, and then only need to load the primary catalog partition for the corresponding HEALPix tile. This greatly speeds up and facilitates the query, as we now only have to load a single tile instead of going through the whole catalog. These are general secondary indexes, and can be built on any column, not just the survey-assigned identifier.

An index table has a different directory structure, as the data is no longer partitioned by HEALPix pixels, at any order. Instead, we have a flat storage of the parquet files, as shown in Figure 7.

3.2.3 Map Catalog

TODO: we should mention this, as it's very useful for science.

3.2.4 Catalog Collection

Margins and indexes are associated with a single astronomical dataset, and to make this connection clearer and enable friendlier application behavior,

```
id_index/
|-- properties
+-- dataset/
    |-- _metadata
    |-- _common_metadata
    +-- index/
        |-- part.0.parquet
        |-- part.1.parquet
        |-- . . .
```

Figure 7: Example index table directory contents

```
gaia_dr3/
|-- collection.properties
|-- catalog/
|   |-- properties
|   +-- . . .
|-- [OPTIONAL] margin_5arcs/
|   |-- properties
|   +-- . . .
|-- [OPTIONAL] margin_10arcs/
|   |-- properties
|   +-- . . .
+-- [OPTIONAL] index_designation/
    |-- properties
    +-- . . .
```

Figure 8: Example collection directory contents

tables MAY be grouped together under a single directory called a catalog collection. These include the primary data `catalog` and other `catalogs` that are optional and are intended either to improve access to the main `catalog` or to enrich it with additional information.

In Figure 8, we present an overview of this folder structure, including a few common examples of such optional catalogs. The catalog collection directory MUST contain a `collection.properties` file, whose content is outlined in Section 3.3.3.

3.2.5 Linked Tables

Noting again that cross-matching and multi-survey analysis is a primary motivator for the HATS spatial format, we introduce linked tables.

This mirrors the relational database notion of two tables that share a "join key" in both tables, and they can be joined at runtime to create a wider table. For each link, there is a single "primary" catalog that reflects the left side, and the other catalog is the "join" catalog (as the data can be joined into the primary catalog).

The linked table should be a valid standalone catalog: the link is merely

a convenience to users who will wish to perform this join, and won't need to know specifics of each table to do so.

Example: Association Table This table provides a spatially-sharded set of links between rows in two different object catalogs. This mirrors the relational database notion of a link table, and will likely be composed of the primary survey identifiers from either side of the association. We make no prescription as to whether the relationships are one-to-one or one-to-many.

The resulting association data will use the coordinates of the primary catalog, and MAY be partitioned with the same pixels as the primary catalog. If there are many more or many fewer association links than there are objects in the primary table, it may be preferable to either combine or split the leaf Parquet files further.

Example: Extension Table This table provides supplemental information that is not stored in the same leaf parquet files of the primary catalog.

Some real-world examples where this is useful:

- **Very wide catalog tables:** For a catalog table with more than 500 columns, it's likely that most scientists will only query or retrieve a small fraction of those columns. Archives and other data providers may wish to move the rarely-used fields to an extension table, that can reside in cheaper storage regimes.
- **Value-add catalog distribution:** Data products created after a survey release, or by other parties, would not be written back to the original catalog data structure. A link provides a convenient way to attach the value-add products back to the survey data for joint analysis.

3.3 Properties

HATS implementations utilize auxiliary files and metadata files to store relevant information about the structure. We make use of a text-based properties file to store general configuration information, and a vocabulary convention for the keys contained in the files.

3.3.1 Format

All `.properties` files use the same format.

It MUST be encoded in UTF-8, with one line per property, following the syntax `keyword = value`. The ordering of the keywords is not important. The keywords MAY include many of those listed in Table 1.

The text file may contain comment lines, beginning with the `'#'` character. Spaces between keyword, equal sign, and value are ignored, but can make the file more human-readable.

We enforce additional requirements for the presence of particular fields for different types of HATS catalogs or tables. A matrix of these requirements is shown in Table 2.

HATS Keyword	**Description - Format - Example**
addendum_description	If content has been added after initial catalog creation, description of any added data - Format: free text
addendum_did	If content has been added after initial catalog creation, creator_did of any added data
all_indexes	For catalog collections, space-delimited map of indexed field to subdirectories containing index tables.
all_margins	For catalog collections, space-delimited list of subdirectories containing margin caches.
bib_reference	Bibliographic reference
bib_reference_url	URL to bibliographic reference
creator_did	Unique ID of the HATS - Format: IVOID - Ex : ivo://CDS/P/2MASS/J
data_ucd	UCD describing data contents (Preite Martinez et al., 2024)
dataproduct_type	Format: one word ONE OF(object, margin, link, index, collection, ...)
default_index	For catalog collections, the field of the default index to use for ID searches.
default_margin	For catalog collections, the subdirectory containing the default margin cache to use for cross-matching
hats_link_join_table_url	For table links, there will be a join table with original survey data (right side of the join)
hats_link_leaf_files	For table links, does the table contain leaf files (may optionally only provide a "soft" link between tiles).
hats_builder	Name and version of the tool used for building the HATS. Format: free text – Example "hats-import v0.6.4"
hats_col_link_join	For table links, column name for the joining (right) side of the join within the original table
hats_col_link_join_link	For table links, column name in the assocation table for the join (right) side of the link.
hats_col_link_primary	For table links, column name for the primary (left) side of the join
hats_col_link_primary_link	For table links, column name in the table that matches the primary (left) side of the join
hats_col_healpix	Column name that provides a spatial index of HEALPix values at some fixed, high order
hats_col_healpix_order	For the `hats_col_healpix` column, what is the HEALPix order
hats_col_dec	Column name of the dec coordinate. Used for partitioning and default cross-matching.
hats_col_ra	Column name of the ra coordinate. Used for partitioning and default cross-matching.
hats_cols_default	Which columns should be read from Parquet files, when user doesn't otherwise specify. Useful for wide tables. Format: space-delimited column names

HATS Keyword	Description - Format - Example
hats_cols_sort	Space-delimited list of the columns used to sort the data. If this isn't specified, it is assumed to be sorted by `hats_col_healpix` column (if present).
hats_cols_survey_id	The primary key used in the original survey data. May be multiple columns if the survey uses a composite key (e.g. object ID and MJD for detections)
hats_coordinate_epoch	For the default ra and dec (hats_col_ra, hats_col_dec), the measurement epoch
hats_copyright	Copyright mention associated to the dataset or HATS - Format: free text
hats_creation_date	HATS first creation date - Format: ISO 8601 => YYYY-mm-ddTHH:MMZ
hats_creator	Institute or person who built the HATS. Format: free text. Ex : CDS (T. Boch)
hats_estsize	HATS size estimation. Format: positive integer. Unit : KB
hats_frame	Coordinate frame reference. Format: word "equatorial" (ICRS), "galactic", "ecliptic"
hats_index_column	For index tables, the column that is indexed over
hats_index_extra_column	For index tables, extra columns that are carried through with the index
hats_margin_threshold	For margin tables, the threshold used for finding points within margin. Units: arcseconds
hats_max_rows	At catalog creation time, the maximum number of rows per file before breaking into 4 new files at higher order.
hats_nrows	Number of rows of the HATS catalog. Format: positive integer
hats_order	Deepest HATS order. Format: positive integer
hats_primary_table_url	For supplemental tables, there will be a primary table with original survey data. Format: URL
hats_progenitor_url	URL to an associated progenitor HATS catalog. Format: URL
hats_release_date	Last HATS update date - Format: ISO 8601 => YYYY-mm-ddTHH:MMZ
hats_service_url	HATS access url. Format: URL
hats_skymap_order	HEALPix order of the default `'skymap.fits'` file
hats_skymap_alt_orders	space-delimited list of HEALPix orders for additional alternate skymap files.
hats_status	HATS status. Format: list of space-delimited words ("private" or "public"), ("main", "mirror", or "partial"), ("clonable", "unclonable" or "clonableOnce"). Default : public main clonableOnce
hats_successor_description	If this HATS should be replaced with a newer catalog, description of replacement or reasoning. Format: free text
hats_successor_url	URL to a newer HATS that replaces this catalog. Format: URL
hats_version	Number of HATS version. Format: 0.1 (corresponds to version of this note)
moc_sky_fraction	Fraction of the sky covered by the MOC associated to the HATS. Format: real between 0 and 1

HATS Keyword	Description - Format - Example
npix_suffix	String to indicate file suffix for leaf files. In the typical HATS directory structure, this is `'.parquet'` or `'.pq'` because there is a single file in each Npix partition. If using leaf directories, `'/'`.
obs_ack	Acknowledgment.
obs_collection	Short name of original data set. Format: one word. Ex : 2MASS
obs_copyright	Copyright associated to the original data. Format: free text
obs_copyright_url	URL to a copyright
obs_description	Data set description. Format: free text, longer free text description of the dataset
obs_regime	General wavelength. Format: word: "Radio" | "Millimeter" | "Infrared" | "Optical" | "UV" | "EUV" | "X-ray" | "Gamma-ray"
obs_title	Data set title. Format: free text, one line. Ex : HST F110W observations
product_type_served	If the HATS contains rich tabular data, specify the semantics of the data. Format: word: "spectra" | "timeseries" | "image"
prov_progenitor	Provenance of the original data. Format: free text
publisher_id	Unique ID of the HATS publisher. Format: IVOID - Ex : ivo://CDS
t_max	Stop time of the observations. Format: real. Representation: MJD
t_min	Start time of the observations. Format: real. Representation: MJD

Table 1: Available keys for properties file

3.3.2 hats.properties

A text file named `hats.properties` is REQUIRED in the root level of the catalog directory. It marks the directory as containing a HATS catalog, and so MUST be located in the root directory of the catalog.

This is also REQUIRED for margin and index tables.

An example `hats.properties` file is shown in Figure 9.

3.3.3 collection.properties

A text file named `collection.properties` is REQUIRED for a catalog collection. It marks the directory as containing a catalog collection, and so MUST be located in the root of the catalog collection.

The keywords MAY include many of those listed in Table 1, with collection-specific fields shown in Table 2.

An example `collection.properties` file is shown in Figure 10.

HATS Keyword	HATS Catalog Type				
	object	margin	index	link	collection
all_indexes					opt
all_margins					opt
dataproduct_type	REQ	REQ	REQ	REQ	
default_index					opt
default_margin					opt
hats_link_join_table_url				REQ	
hats_link_leaf_files				REQ	
hats_col_link_join				REQ	
hats_col_link_join_link				opt	
hats_col_link_primary				REQ	
hats_col_link_primary_link				opt	
hats_col_dec	REQ	opt			
hats_col_ra	REQ	opt			
hats_cols_default	opt	opt			
hats_index_column			REQ		
hats_index_extra_column			opt		
hats_margin_threshold		REQ			
hats_npix_suffix	opt	opt	opt	opt	
hats_nrows	REQ	REQ	REQ	REQ	
hats_primary_table_url		REQ	REQ	REQ	REQ
obs_collection	REQ	REQ	REQ	REQ	REQ

Table 2: Catalog-type specific fields. For display, REQ is REQUIRED, and opt is OPTIONAL

```
#HATS catalog
obs_collection     = euclid_q1_merFinalCatalog
dataproduct_type   = object
hats_nrows         = 29767806
hats_col_ra        = RIGHT_ASCENSION
hats_col_dec       = DECLINATION
hats_cols_sort     = OBJECT_ID
hats_max_rows      = 1000000
hats_order         = 6
moc_sky_fraction   = 0.00618
hats_builder       = hats-import v0.4.4
hats_creation_date = 2025-03-20T03:38UTC
hats_estsize       = 23137775
hats_release_date  = 2024-09-18
hats_version       = v1.0
```

Figure 9: Example `hats.properties` file contents

3.3.4 Linked table properties

3.4 Metadata and Auxiliary Files

A HATS catalog may contain other arbitrary files in the catalog root directory level, as desired by the catalog publisher, or necessary for data access. A

```
#HATS Collection
obs_collection=gaia_dr3
hats_primary_table_url=gaia
all_margins=gaia_5arcs gaia_1arcs
default_margin=gaia_5arcs
all_indexes=designation designation_index
```

Figure 10: Example `collection.properties` file contents

```
Norder,Npix
3,530
4,637
4,958
4,1003
4,2147
```

Figure 11: Example `partition_info.csv` file contents

HATS reader MUST allow for unexpected files at the catalog root directory level.

Note that collection.properties is only required if creating a `catalog` collection for your dataset.

We will now go over these files and explain their function, format and their contents.

3.4.1 partition_info.csv

A text file named `partition_info.csv` is OPTIONAL and RECOMMENDED in the root level of the catalog directory. If present, it MUST be a CSV (comma-separated-values) file, with the columns `"Norder"` and `"Npix"`, as shown in example contents in Figure 11. Additional columns might be present in the file, but are not required, and may not be interpreted by all HATS readers. The values of pairs of `"Norder"` and `"Npix"` reflect the HEALPix tiles of the catalog's partitions. HATS readers can quickly read this file to understand the full scope of the catalog, and potential spatial overlap with other catalogs.

3.4.2 skymap.fits

A FITS file containing the HEALPix skymap of the catalog. This is a two-dimensional histogram of points in each HEALPix tile at some reasonably high order. This will be at the highest order calculated during the catalog ingestion process (likely 9 or 10). This data is useful when inspecting catalogs and understanding the distribution of data.

This file is OPTIONAL and RECOMMENDED for object catalogs.

```
hats_skymap_order      = 8
hats_skymap_alt_orders = 2 4 6
hats_order             = 6
```

(a) Example properties for the HEALPix order of skymaps

```
skymap.2.fits
skymap.4.fits
skymap.6.fits
skymap.fits
```

(b) Example skymaps in the root directory of the catalog

Figure 12: Example catalog with down-sampled skymaps

HATS clients may find a `skymap.fits` file that has been generated at HEALPix order 9 to be very large, and more detail than they need. Down-sampled skymaps can provide a more manageable file size while still offering a useful overview of the data distribution. These skymaps can be generated at lower HEALPix orders, allowing for quicker visualizations and analyses without the need for the full-resolution data.

Additional skymaps are named `skymap.<healpix order>.fits`, within the same root directory of the catalog as the `skymap.fits` file, and are listed in the `hats.properties` file.

Example snippets are shown in Figure 12. For the example, we calculated the skymap at order 8 during catalog ingestion, and have written this to the `skymap.fits` file. Based on the data distribution, we have a highest HEALPix order of data partitions of 6. The catalog provider has downsampled the skymap to orders 2, 4, and 6. We expect this catalog to have 4 skymap files at the root level: the primary and default at order 8, and 3 additional down-sampled files.

3.4.3 Images and static summaries

Catalog providers may wish to provide human users with footprint previews and data summaries. All such files are OPTIONAL, but we enumerate some of them here to provide motivation, and a framework for consistent naming. This is not meant to be exhaustive; remember that a HATS reader MUST allow for unexpected files at the catalog root directory level.

skymap.png A file called `skymap.png` MAY be provided in the catalog root to offer a preview of the distribution of data on the sphere. It should match the distribution of data provided in the `skymap.fits`.

This MAY also be included in the collection root directory, and should reflect the data of the collection's primary catalog.

No required pixel size or display format is specified.

partition_info.png A file called `partition_info.png` MAY be provided to offer a preview of the relative density of HATS partitions. This should

show the HEALPix order used for partitioning, across the footprint of the catalog data.

Figure 4 is one such example of plotting partition info.

No required pixel size or display format is specified.

README A file called `README.md`, `README`, or `README.txt` MAY be provided to offer a human-readable summary of the HATS catalog. This MAY contain information such as sky coverage, embed a `skymap.png`, information on auxiliary tables (e.g. margins), columns, types, provenance, acknowledgments, or more user-level information such as accessing the catalog with common tools. The internal format and content of this file is intentionally not specified.

This MAY also be included in the collection root directory.

index.html A file called `index.html` MAY be provided to offer an HTML summary of the HATS catalog, displayable through any web browser. This MAY contain information such as sky coverage, embed a `skymap.png`, information on auxiliary tables (e.g. margins), columns, types, provenance, or more user-level information such as accessing the catalog with common tools. The internal format and content of this file is intentionally not specified; it is appropriate that this file be styled to suit the catalog provider, and MAY contain any additional links to help the user access and understand the data.

This MAY also be included in the collection root directory.

3.4.4 Additional Parquet Files

There are several ways to re-structure the parquet data in ways that compress or summarize the available data. All such files are OPTIONAL, but we enumerate some of them here to provide motivation, and a framework for consistent naming. This is not meant to be exhaustive; remember that a HATS reader MUST allow for unexpected files at the catalog root directory level.

data_thumbnail.parquet This is a small dataset aimed to help users to understand and use the data. It MUST have the same overall data schema as the data partitions themselves. It MUST not be larger than the threshold used to split data partitions. The data it contains should represent the diversity of the catalog well.

This file gives the user a quick overview of the whole dataset. Given how it is sampled, it will cover the entire width of the dataset, and give a reasonably accurate overview of the properties of the dataset. It is thus both more convenient than, and superior to, directing a user to a subset of any single Parquet data partition for the same purposes.

This file is OPTIONAL and RECOMMENDED for object catalogs.

_metadata and _common_metadata Many Parquet reading frameworks support and recommend additional dataset-level metadata files:

- `_common_metadata` which contains the full schema of the dataset. This file will know all of the columns and their types, as well as any file-level key-value metadata associated with the full Parquet dataset.
- `_metadata` contains per-partition information, chiefly the storage information and column metadata and statistics.

Both files are OPTIONAL and RECOMMENDED for all catalogs.

To understand the importance of these files, it is helpful to understand the structure of partitioned Parquet files. Figure 13 shows a schematic of two partitioned Parquet files.

The data values are shown in gray, and typically will take up most of the space of the Parquet file. Data is stored and compressed on disk per "row group". File 1 in Figure 13 contains less data, and it all fits into a single row group. File 2 in Figure 13 contains more data, and is split on disk into two row groups. All of the metadata is stored inside the file footer. There is some file-level metadata that may describe the full dataset, as well as column-level key-value metadata. Parquet files MAY also have aggregate statistics about values in each column (minimum value, maximum value, count of valid values).

For very large datasets, it is helpful to have a single file that holds the common data schema information in all of the partitioned Parquet files (assuming that they have homogeneous structure). This `_common_metadata` file can be much smaller (and so faster to read) than a leaf Parquet file. The column statistics, however, will be different for every file, and some files may contain multiple statistics segments if the data inside a single file is very large and has been divided into multiple row groups. These statistics can be concatenated into a single `_metadata` file, and can provide valuable insight into the distribution of the data. A clever Parquet reader can use this information to filter queries to only those partitions where certain values are possible. See Figure 14 for the layout of these two Parquet metadata files.

per_partition_statistics.parquet This is a subset of the data contained in the `_metadata`, and provides basic statistics of all values inside the data partitions. This file is OPTIONAL and RECOMMENDED for object catalogs.

File 1

data				row group 1
footer	file-level key-value metadata			
	id - type - key-value metadata	ra - type - key-value metadata	dec - type - key-value metadata	column metadata
	stats min, max, count	stats min, max, count	stats min, max, count	stats for row group 1

File 2

data				row group 1
data				row group 2
footer	file-level key-value metadata			
	id - type - key-value metadata	ra - type - key-value metadata	dec - type - key-value metadata	column metadata
	stats min, max, count	stats min, max, count	stats min, max, count	stats for row group 1
	stats min, max, count	stats min, max, count	stats min, max, count	stats for row group 2

Figure 13: Example file layout of two Parquet files of a dataset

_common_metadata

file-level key-value metadata		
id - type - key-value metadata	ra - type - key-value metadata	dec - type - key-value metadata

_metadata

file-level key-value metadata			
id - type - key-value metadata	ra - type - key-value metadata	dec - type - key-value metadata	column metadata
stats min, max, count	stats min, max, count	stats min, max, count	stats for File 1, row group 1
stats min, max, count	stats min, max, count	stats min, max, count	stats for File 2, row group 1
stats min, max, count	stats min, max, count	stats min, max, count	stats for File 2, row group 2

Figure 14: Example file layout of two supplemental Parquet files of a dataset

4 Performance Considerations

Here, we will elaborate on several ways in which this format can be efficiently used. These insights come from our work with LSDB[3], a Python implementation of a package that works natively with HATS catalogs.

4.1 Parquet Storage

Firstly, we emphasize the need to use the Parquet column filtering. Loading into memory only the columns that a user needs for scientific analysis, typically just a few out of the tens or hundreds available, significantly reduces the computational requirements for the analysis.

We can also split Parquet files into so-called row groups, splitting Parquet into chunks with a fixed number of rows. Parquet readers can skip over entire row groups if they don't contain relevant data, using the Parquet row group footer's min and max values for hints. This is especially effective when row groups are designed to match the access patterns of a specific scientific use case. For instance, if rowgroups are made to be small and sorted by the identification number of the survey, the retrieval of the individual rows by survey identification (discussed in Section 3.2.2) can be made much faster. Speed-up happens because we don't have to load the entire Parquet file into memory, but only the smaller rowgroup, to retrieve the needed row with the required identification number.

4.2 HTTP Services

The fact that the data can be stored on the hard drive and served to the users simplifies the cost structure for catalog providers. However, a user operating on the dataset, even if they are doing aggressive filtering and requesting a minimal number of rows at the end, will still have to transfer a large amount of data to their client, where the filtering is actually conducted. These limitations could become prohibitive if done over a network or with limited bandwidth. To alleviate that problem, it is possible to implement a server-side query in which the filtering operations are done server-side, and only the final dataset is sent to a user. Of course, this requires computational resources on the provider's side, but operates on a single file and will benefit from Parquet storage optimizations from Section 4.1

4.3 Cross-matching

Finally, we want to highlight the exceptional performance possible when cross-matching HATS catalogs. Due to its spatial sharding, the cross-matching approach implemented in LSDB is competitive with the existing tools for

[3] https://lsdb.io

small data, and is more efficient for large catalogs, starting with roughly one million rows. The key advantage lies in the fine-grained spatial partitioning: each data chunk is designed to fit comfortably into memory, allowing efficient parallel processing without memory bottlenecks. Users can increase the number of parallel computation units, and—provided the number of workers remains below the number of partitions and I/O throughput is sufficient—achieve near-linear scaling.

For typical cases of large catalogs (more than billion rows), cross-matching on a single core is around 5 to 15% slower than the pure I/O speed. As discussed above, selecting only specific columns and parallelizing the work can drastically improve performance.

5 Operational Considerations

5.1 Provenance

5.2 Mirroring and versioning

5.3 Validation

5.4 Registry

6 Role within the VO Architecture

Figure 15 shows the role this document plays within the IVOA architecture.

6.1 HiPS

HATS is inspired by the HiPS standard (Fernique et al., 2017). HATS shares similar principles and structure, but serves a different purpose. This section briefly tackles the similarities and differences in the approaches, as motivated by the distinct use cases.

Both HATS and HiPS are based on the hierarchical nature of the HEALPix NESTED indexing strategy. Both are designed to serve static files in a directory structure, possibly behind a minimalist web server. Property files are made of a simple list of key/value pairs. They are easy to parse and to interpret, e.g. as Java properties files. HiPS and HATS both rely on well supported – and even standardized – HEALPix files: MOC (Fernique et al., 2019) for HiPS , HEALPix skymap for HATS.

Here we discuss the two main differences.

1. First, HiPS is primarily made for spatial visualisation with the idea that "the more you zoom, the more you get". The mechanism includes row resolution information, and all HEALPix NESTED layers contain

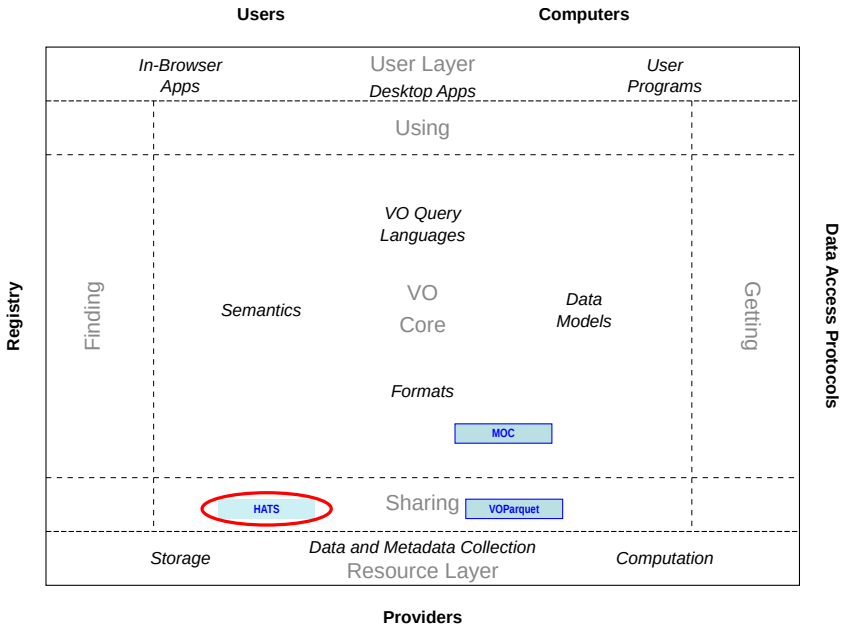


Figure 15: Architecture diagram for this document

(possibly degraded or partial) information. This means some information about a tile at a given depth is available in parent cells. Moreover, each tile (including leaf tiles) is usually quite small (a few tens to a few hundreds of kilobytes).

On the contrary, HATS decribes an adaptive grid with information distributed in the HEALPix NESTED tree leaves, without partial or redundant information on nodes. Each leaf is usually quite large (typically a few hundreds of megabytes). The main goal is the possibility to perform operations (including row-filtering) on sub-columns, on the whole dataset (often meaning the whole sky).

2. Second, HiPS does not rely on any given external implementation. The implementer of a client supporting the HiPS standard has to write its own, moderately complex, code. The moderate size of a HEALPix tile is guaranteed by construction in the standard.

   On the other hand, HATS utilizes existing industry standards and thus benefits from the external implementation of code, as well as ongoing community maintenance of the software. The possible reduction of the amount of data retrieved per tile depends on both the request and the implementations (assuming static files).

There are several formal similarities between HiPS and HATS. At the root level, both:

- require a key/value pairs properties file, with partially overlapping lists of keys
- recommend to have coverage information
  - HiPS: `MOC.fits`
  - HATS: `partition_info.csv` (recommended) and `skymap.fits` (density map, optional)
- allow for an optional preview:
  - HiPS: `preview.jpg`
  - HATS: `data_thumbnail.parquet`
- the directory structure of HEALPix tiles is similar (the '=' in HATS is required by parquet readers):
  - HiPS: `/NorderK/DirD/NpixN`
  - HATS: `/Norder=K/Dir=D/Npix=N`
  - in both, `DirN` is made to pack groups of 10000 rows

To sum up, HiPS and HATS both leverage the HEALPix hierarchical nature and translate it into a nested directory structure. The HiPS standard is designed primarily for progressive visualization, with each HEALPix order containing information. HATS format optimizes operations on (and between) catalogs and stores information at the most precise order without redundancy. These two distinct goals justify the need for distinct standards.

6.2 VOParquet

The VOParquet format (Taylor et al., 2025) specifies that the VO metadata should be present in the file-level key-value metadata of the parquet file. We RECOMMEND that the VOParquet metadata be included in all parquet files, be they ancilliary metadata files (`_common_metadata`, `_metadata`, `data_thumbnail.parquet`) or data partitions. If VOParquet metadata exists in one data partition file, it MUST be present in all data partitions.

HATS catalogs MAY contain additional parquet files beyond the data partitions, as discussed in Section 3.4. The mix of optional files and optional metadata within the files will introduce points of confusion. Here we discuss the motivation behind our recommendations, and their consequences.

`_common_metadata` contains the merged schema for all data partition parquet files. If a `_common_metadata` file exists, it WILL have a superset of all key-value metadata present in the data partitions. It follows that it MUST have the VO metadata present to be considered VOParquet.

`_metadata` contains some schema information, but is not required to contain all schema information. It MAY contain the VOParquet metadata, but if it does NOT contain the metadata, the catalog may still be considered valid VOParquet if the metadata is present elsewhere.

`data_thumbnail.parquet` MUST have the same overall data schema as the data partitions themselves. It follows that it MUST have the VO metadata present to be considered VOParquet.

If the `_common_metadata` and `data_thumbnail.parquet` files are not present, the VO metadata must be present in ALL data partitions to be considered VOParquet.

6.3 Compatibility with other standards

As discussed above, HATS is designed to be closely integrated with existing VO spatial indexing frameworks, such as HEALPix and MOC (Multi-Order Coverage maps). It builds upon the nested HEALPix tessellation scheme, which provides a hierarchical and uniform way to partition the celestial sphere.

The HATS format can be made to be compatible with the TAP query (Dowler et al., 2019) by implementing a translation layer between the TAP query language. We have explored some initial implementation of such functionality, but the implementation details will always depend on the language used to handle the Parquet files.

A Changes from Previous Versions

No previous versions yet.

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